MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2015
Higher 2

CANDIDATE NAME

CIVICS GROUP	1	4	S			

INDEX NUMBER

H2 BIOLOGY 9648/02

Paper 2 Core Paper

17 September 2015
2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A
Answer **all** questions in the spaces provided on the question paper.

Section B
Answer **one** question on the answer paper provided.

At the end of the examination,
1. Fasten your answer papers to section B securely together.
2. Hand in the following separately:
   - Section A (Part I)
   - Section A (Part II)
   - Section B

The number of marks is given in brackets [ ] at the end of each question or part question.

For examiner's Use	
Section A	
1	/ 15
2	/ 10
3	/ 12
4	/ 8
5	/ 12
6	/ 10
7	/ 13
Section B	
8 / 9	/ 20
Total	**/ 100**

This paper consists of **18** printed pages and **2** blank pages.

[Turn over]

Section A (Part I)

Answer **all** the questions in this section.

QUESTION 1

Protein function depends on the specific shape of the newly synthesized protein. The tertiary structure of globular proteins depends on the way they are folded.

In the cytosol, chaperone proteins can bind reversibly to newly synthesized polypeptides and shield them from other molecules. This prevents the R groups of amino acids from forming bonds with the wrong molecular partners, hence allowing the polypeptides to fold correctly.

Fig. 1.1 below shows how chaperone proteins work.

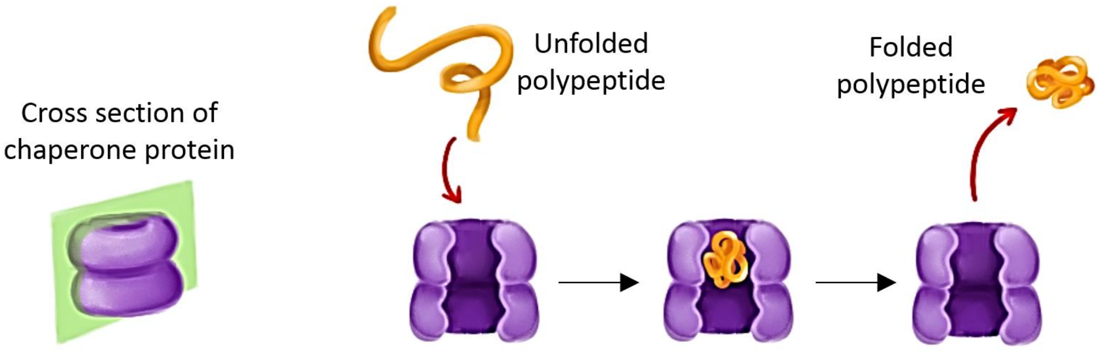

Fig. 1.1

a) With reference to Fig. 1.1, explain how chaperone proteins regulate gene expression. [2]

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b) Newly formed polypeptides, together with their chaperones, must be moved from the site of synthesis to their target location in the cell, where they adopt their final shape to carry out their specific function.

Suggest one reason why this movement is unlikely to be by diffusion. [1]

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For Examiner's Use

When a protein fails to fold correctly upon synthesis or misfolds at a later stage in its cellular life time, it can no longer fulfil its biological function. This may happen as a result of either:

- Ageing, which is associated with a decrease in expression of chaperone proteins, or
- An inherited mutation in a gene, which causes a change in the amino acid sequence of a protein.

Protein misfolding can result in diseases because some misfolded proteins tend to aggregate into fibrils, which may result in cell death.

Fig. 1.2 shows how misfolded proteins aggregate.

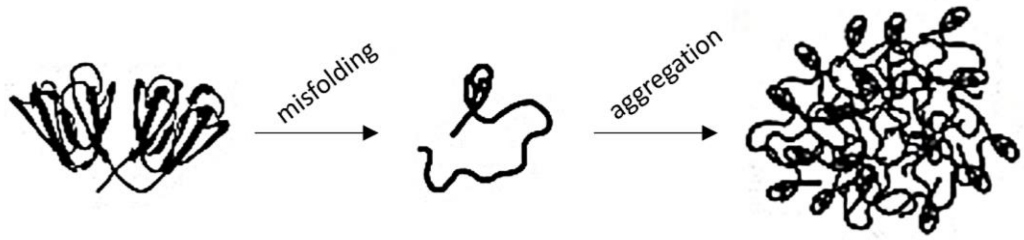

Fig. 1.2

c) Describe how cytosolic proteins are broken down in cells. [2]

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d) With reference to Fig. 1.2, suggest why misfolded proteins tend to aggregate in the cell. [2]

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An example of a misfolding disease is Alzheimer's disease, a neurodegenerative disease. In Alzheimer's disease, there is an accumulation of misfolded amyloid β protein in the neurons of patients, leading to symptoms of the disease.

For Examiner's Use

Hereditary forms of Alzheimer's disease have been associated with a mutation in the amyloid β precursor protein (APP) gene, which encodes the APP protein. The APP protein is a transmembrane glycoprotein found on the surface of many cells, such as neurons. The proteolysis of the APP by enzymes produces the amyloid β protein.

e) Describe how the APP protein on the cell surface membrane is formed after its polypeptide chain has been synthesized. [4]

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One mutation on the APP gene involves one codon being changed from GCC to GUC. Fig. 1.3 shows the genetic code.

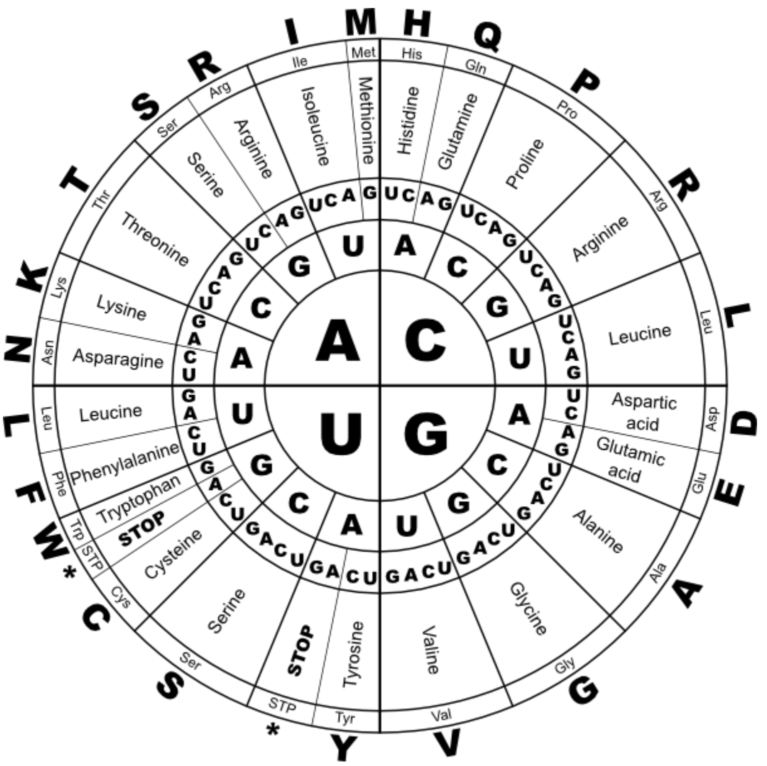

Fig. 1.3

For Examiner's Use

f) Explain the term *codon*. [2]

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g) With reference to Fig. 1.3, describe the mutation that has occurred. [2]

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[Total: 15]

QUESTION 2

In normal cells, the cell cycle is controlled at checkpoints. These checkpoints involve proteins called cyclin-dependent-kinases (CDKs). CDKs are inactive, but can be activated by binding to cyclins to form cyclin-CDK complexes. Active CDKs activate target proteins that lead to cell cycle progression.

Different cyclin-CDK complexes trigger different steps in the cell cycle.

Fig. 2.1 shows how the formation of one type of cyclin-CDK complex triggers mitosis.

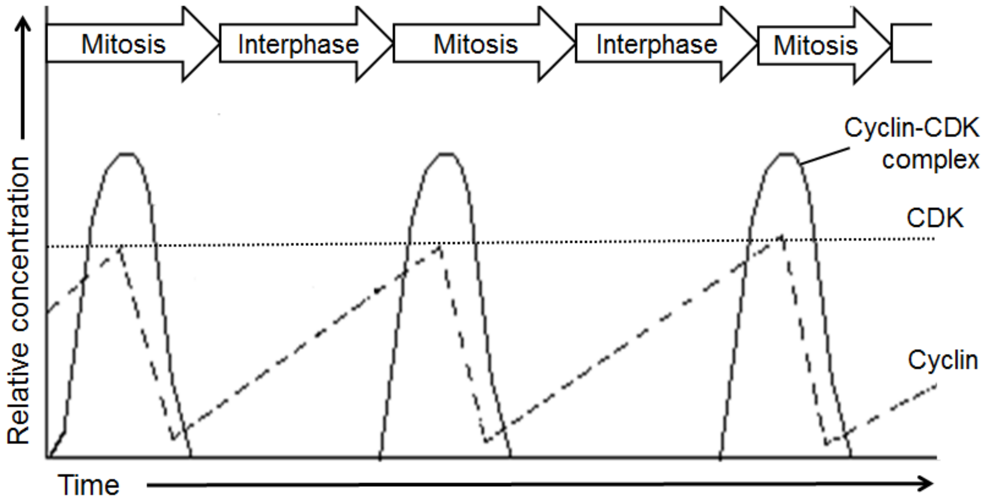

Fig. 2.1

a) With reference to Fig. 2.1, explain how the levels of cyclin-CDK complexes fluctuate. [2]

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b) Describe how target proteins can be activated by active CDKs. [2]

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For Examiner's Use

For Examiner's Use

c) Describe the roles of the checkpoints that control the cell cycle. [3]

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d) Explain how the dysregulation of checkpoints may lead to cancer. [3]

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[Total: 10]

For Examiner's Use

QUESTION 3

Fig. 3.1 shows the methylation pattern of a segment of DNA from a human chromosome taken from three different cell types.

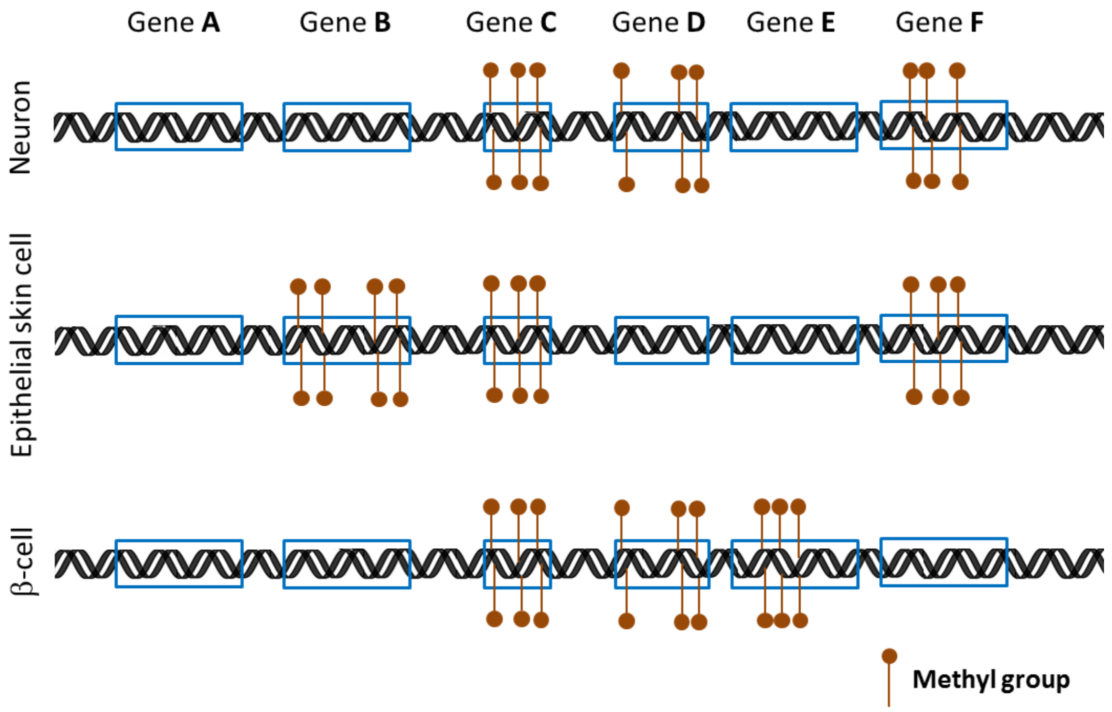

Fig. 3.1

a) Explain the effect of DNA methylation on gene expression. [4]

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b) Explain why the pattern of DNA methylation differs in the three cell types. [3]

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For Examiner's Use

c) Suggest why gene **A** is not methylated in all the cell types and hence suggest a possible identity of gene **A**. [2]

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Methylation of DNA always occurs on the cytosine base of a CG dinucleotide. The methylation pattern is heritable when a cell divides by mitosis. Fig. 3.2 illustrates this process.

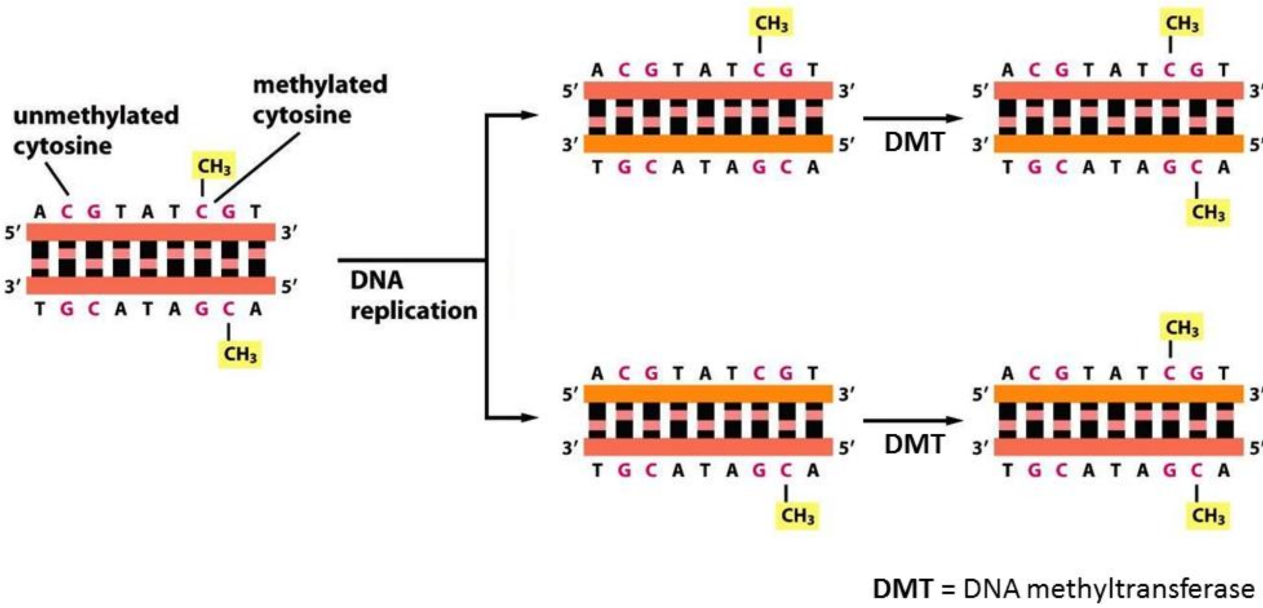

Fig. 3.2

d) Using information in Fig. 3.2, suggest how it is possible for the methylation pattern to be inherited. [3]

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[Total: 12]

For Examiner's Use

QUESTION 4

Blood donated to blood banks undergo stringent HIV testing before being made available to hospitals. Nucleic acid tests have been developed to detect RNA in the HIV particles in the blood plasma and to detect HIV DNA in infected blood cells.

Fig. 4.1 shows the levels of HIV RNA and HIV DNA in the first 50 days of infection.

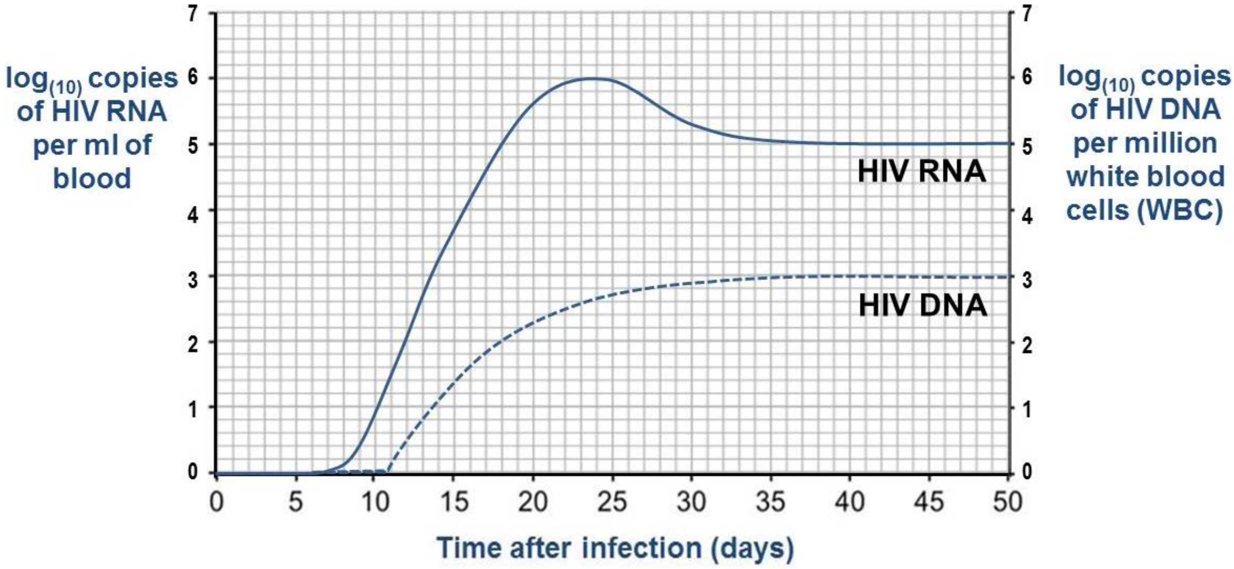

Fig. 4.1

a) State a specific type of white blood cell from which the level of HIV DNA can be measured. [1]

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b) With reference to Fig. 4.1,

i) explain why HIV is considered to be a retrovirus. [3]

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ii) account for the levels of HIV DNA after 35 days of infection. [3]

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For Examiner's Use

c) Antibiotics are often given to HIV/AIDS patients to treat secondary bacterial infections, but not to treat the HIV infection.

Suggest why. [1]

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[Total: 8]

QUESTION 5

For Examiner's Use

The mode of inheritance for shape of summer squash can be investigated by a cross between two summer squash plants with disc-shaped fruit. The cross resulted in offspring with 3 different phenotypes in the following numbers:

Disc-shaped fruit	1071
Sphere-shaped fruit	720
Long fruit	123

Shape of summer squash is determined by 2 gene loci, **A/a** and **B/b**.

a) Explain how different fruit shapes in summer squashes come about. [3]

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b) With this understanding of their mode of inheritance, a breeder crossed two sphere-shaped summer squash plant together. None of the offspring produced gives sphere-shaped fruit.

Construct a genetic diagram to explain the cross. [3]

For Examiner's Use

c) A test cross is used to determine the genotype of an organism.

Comment on the effectiveness of using a test cross to determine the genotype of a disc-shaped squash. [3]

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Squashes also possess different fruit colours. Fruit colours of squash are determined by two other gene loci on 2 different chromosomes. The colours of squash are determined by gene locus **C/c**. The presence of a dominant allele, **C** results in white squash, while the recessive allele gives rise to coloured squash. At the second gene locus **G/g**, allele that codes for yellow is dominant to allele that codes for green.

A cross between two heterozygous white squash plants gives phenotypes in the following numbers:

White squash	250
Yellow squash	60
Green squash	10

The expected ratio for the above cross is 12:3:1.

The χ^2 distribution table and equation to calculate χ^2 is shown below.

number of degrees of freedom (v)	probability
	0.05
1	3.84
2	5.99
3	7.82
4	9.49

$$\chi^2 = \Sigma \frac{(O-E)^2}{E}$$

O = observed result
E = expected result

d) Using the equation and χ^2 distribution table given above, explain if epistasis is the correct explanation for the results obtained. [3]

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[Total: 12]

Candidate Name	Civics Group	Index No.

For Examiner's Use

Meridian Junior College
JC2 Preliminary Examinations 2015
H2 Biology (9648/02)

Question 6	Question 7
/ 10	/ 13

Section A (Part II)
Answer **all** the questions in this section.

QUESTION 6
Contraction of airway smooth muscle is controlled by the release of acetylcholine from presynaptic neurones. Binding of acetylcholine to G-protein coupled receptors (GPCR) embedded in the smooth muscle membrane eventually result in the contraction and growth of smooth muscles. Fig. 6.1 below show part of this signalling pathway.

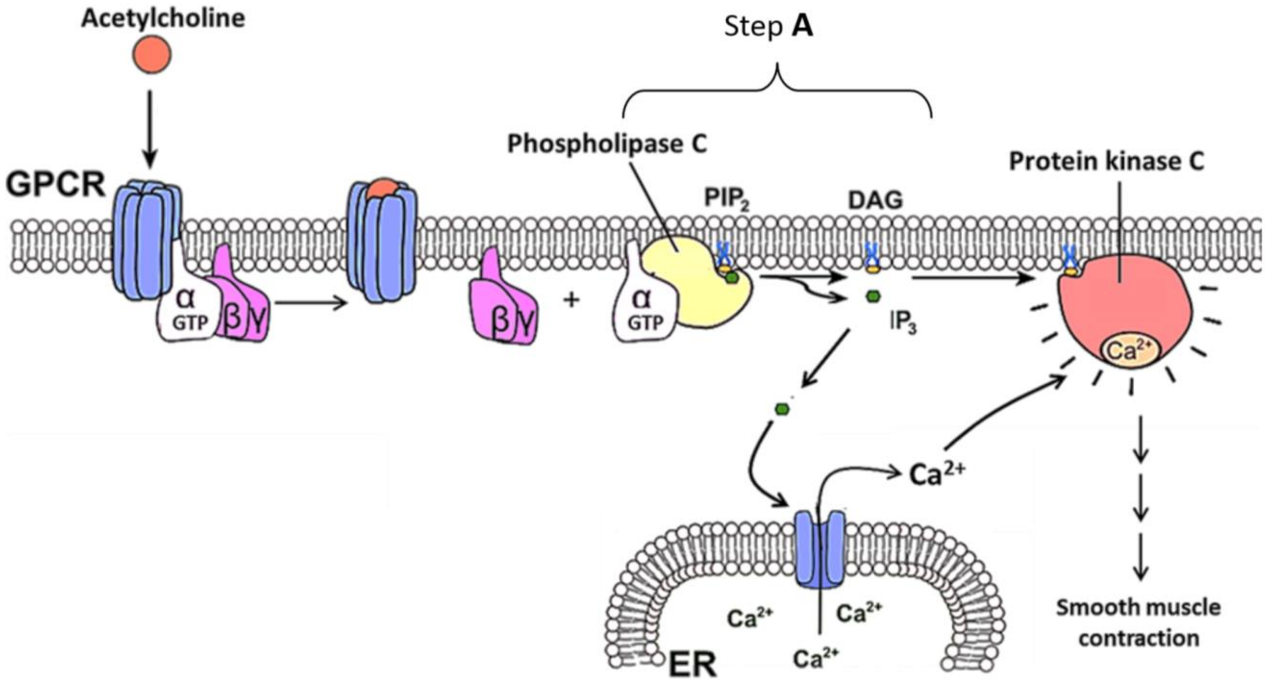

Fig. 6.1

a) With reference to Fig. 6.1,

i) describe one distinctive structural feature of the G-protein coupled receptor. [1]

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ii) explain the role of G-protein coupled receptor. [2]

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iii) explain the significance of step **A** in facilitating an effective transduction. [3]

For Examiner's Use

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b) Describe **one** difference in the roles of Ca^{2+} in triggering contraction of smooth muscles and in the release of acetylcholine in neurons. [2]

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c) A mutation resulted in high concentration of DAG and IP_3 being produced in the muscle cell.

Assuming that the phospholipase C functions normally, explain how a mutation can lead to the observation above. [2]

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[Total: 10]

For Examiner's Use

QUESTION 7

a) Outline the role of variation in evolution. [4]

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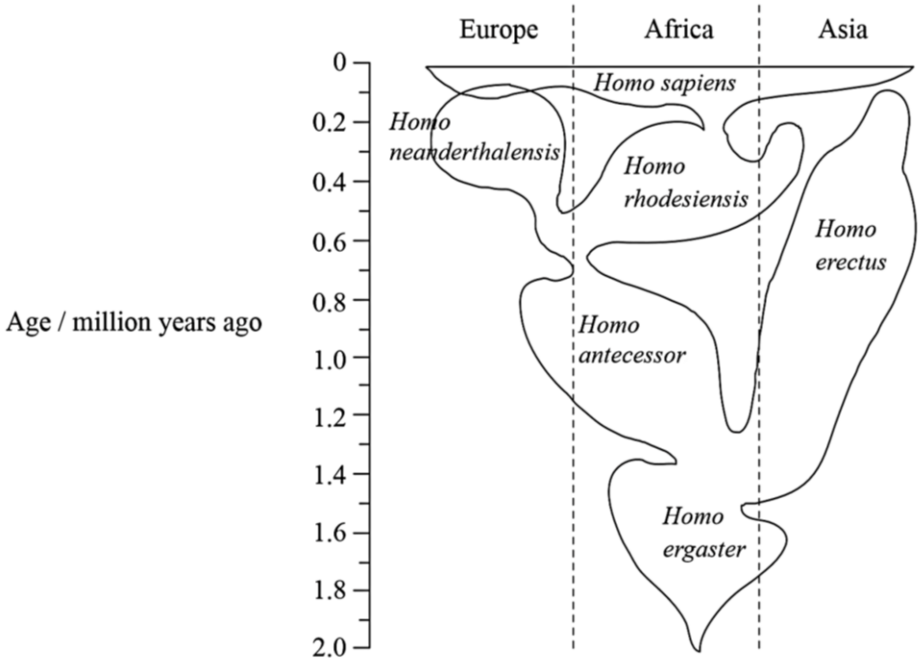
b) Recently discovered fossilized partial skulls from Ethiopia have raised new questions about early human evolution. This has led to different theories about the origins of *Homo sapiens*. One of these theories is illustrated in Fig. 7.1.



[Source: Stringer, Nature (2003), 423, pages 692-695]

Fig. 7.1

i) Identify the species that shows the greatest geographical distribution. [1]

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ii) Suggest, with reason, whether *H. antecessor* or *H. erectus* would likely provide more fossil evidence. [2]

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c) The evolution of groups of living organisms can be studied by comparing the base sequences of their DNA. If a species diverges into two groups, differences in base sequence between the two species accumulate gradually over long periods of time. The number of differences can be used as an evolutionary clock.

Samples of DNA were recently obtained from fossil bones of a Neanderthal (*Homo neanderthalensis*). A section of the DNA from the mitochondrion was chosen for study, as it shows a high level of variation in base sequence between different individuals. A section of the Neanderthal mitochondrial DNA was sequenced and compared with sequences from 994 humans and 16 chimpanzees. The bar chart below shows the number of base sequence differences found among humans, between humans and the Neanderthal, and between humans and chimpanzees.

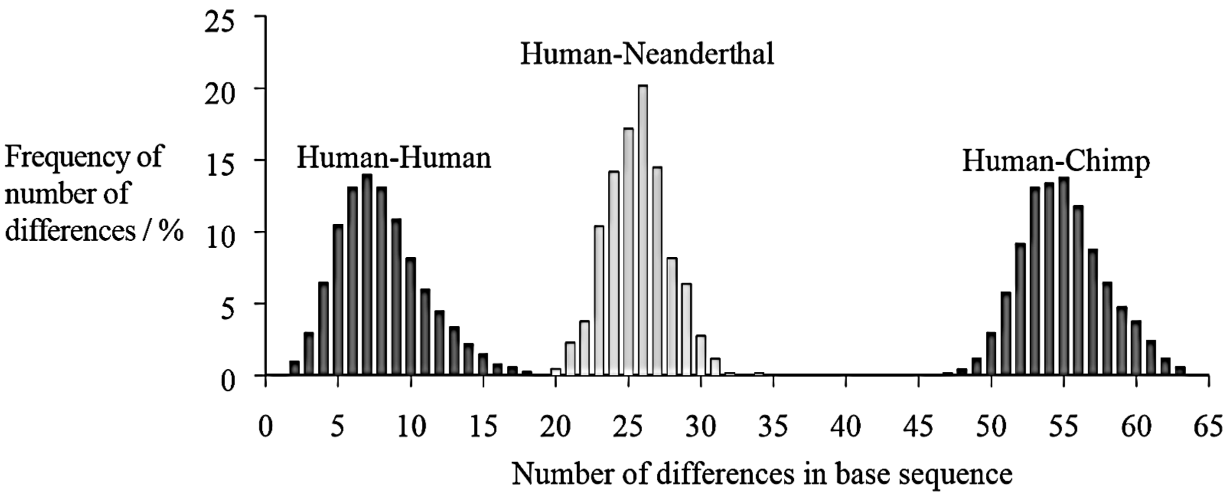

Fig. 7.2

i) The number of base sequence differences between pairs of humans varied from 1 to 18.

State the number of differences shown by the highest percentage of pairs of humans. [1]

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ii) Humans and Neanderthals are both classified in the genus *Homo* and chimpanzees are classified in the genus *Pan*.

Discuss whether this classification is supported by the data in Fig. 7.2. [3]

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For Examiner's Use

For Examiner's Use

d) Explain why the evolutionary clock supports the neutral theory of molecular evolution. [2]

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[Total: 13]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be in set out in questions **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 8

a) Compare tropocollagen and amylose. [7]

b) The optimum pH for the activity of rubisco is pH8 (alkaline). Explain why the illumination of chloroplasts leads to optimum pH conditions for rubisco. [7]

c) Describe the differences in behavior of homologous chromosomes during mitosis and the first division of meiosis. [6]

[Total: 20]

QUESTION 9

a) The cell surface membrane consists of phospholipids and cholesterol. Describe how their structures and properties are related their functions. [8]

b) Describe what happens to pyruvate if yeast is deprived of oxygen. [5]

c) Explain the meaning of the term homeostasis with specific reference to the control of raised blood glucose concentration in mammals. [7]

[Total: 20]

• END OF PAPER 2 •